

Lem aditividad imp homogeneidad racionales

Lema 1. Sea X un espacio vectorial sobre $\mathbb{K} = \mathbb{R}$ o $\mathbb{C}$ y $f: X \rightarrow \mathbb{K}$ aditiva

$\implies f$ es homogénea para escalares racionales $\lambda \in \mathbb{Q}$.

Demostración: Sea $x \in X$ y $\lambda \in \mathbb{Q}$.

Caso I ($\lambda \in \mathbb{N}$): Por aditividad de f , $f(nx) = f(x + \dots + x) = f(x) + \dots + f(x) = nf(x)$.

Caso II ($\lambda \in \mathbb{Z} \setminus \mathbb{N}$): Tenemos que $f(0) = f(0 + 0) = f(0) + f(0) \implies f(0) = 0$. Por tanto, $f(-nx) = f(0 - nx) = f(0) - f(nx) = -nf(x)$ por el Caso ??.

Caso III ($\lambda \in \mathbb{Q}$): Sea $\lambda = \frac{p}{q}$ con $p \in \mathbb{Z}$, $q \in \mathbb{N}$. Entonces:

$$pf(x) \stackrel{\text{Caso ??}}{=} f(px) = f\left(q \cdot \frac{p}{q}x\right) \stackrel{\text{Caso ??}}{=} qf\left(\frac{p}{q}x\right) \implies f\left(\frac{p}{q}x\right) = \frac{p}{q}f(x). \quad \blacksquare$$

Referenciado en

- Lem-aditividad-no-homogenea-imp-grafica-densa
- Lem-aditividad-continuidad-imp-homogeneidad

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